

Agentic vs. Classical In- Market Regime Monitoring

Experimental Setup

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Experimental Setup: Agentic vs. Classical In-Market Regime Monitoring

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Experimental Setup: Agentic vs. Classical In-Market Regime Monitoring

Project: In-market agentic-AI monitoring of quantitative trading strategies **Scope of this document:** the experimental design, data methodology, system architecture, evaluation protocol, and the multi-market extension (US S&P 500 → global energy), including a worked case study of the agentic layer applied to the March-2020 oil crash.

1. Research question and hypotheses

Quantitative trading strategies degrade when the market **regime** they rely on breaks (e.g., a mean-reversion book during a momentum crash). The question:

Can an **LLM-agent monitoring layer**, fed news and macro context under strict information parity, detect strategy regime breaks **faster** and with **fewer false positives** than classical statistical change-point detectors?

Formal hypotheses:

- **H1 — Superiority.** Agentic monitoring detects regime breaks with **lower detection latency** and a **lower false-positive rate** than the classical layer.
- **H2 — Generalisation.** The advantage holds across **both** strategies (a statistical-arbitrage book and a cross-sectional momentum book).
- **H3 — Heterogeneity.** The size of the advantage **differs by strategy**.

The design is a **2 × 2 × N**: {classical, agentic} × {strategy A, strategy B} × {N labelled evaluation windows}.

2. Core methodological principle: information parity (no look-ahead)

The single most important control. **Neither layer may see anything that was not knowable on the decision date.** This is enforced at four levels:

1. **Survivorship-bias-free universe.** Strategy universes are reconstructed point-in-time, *including delisted names*. This is not cosmetic: the August-2007 quant meltdown is **invisible** on a survivor-only S&P 500 (+3.4% in the window) and only appears on the honest universe (−4.8%, worst day −4.7%).
2. **As-of macro data.** Macro series are pulled from **ALFRED vintages** (the value as *published* by the decision date), not the latest revised value — using the revised value would be look-ahead.
3. **Timestamp-filtered news.** Every news article is filtered by its publication timestamp to the decision date.
4. **Fail-closed lookahead assertion.** Every context object passed to an agent is run through a `assert_no_lookahead` guard before the LLM call.

A separate **pre-training-leakage control** (see §8) addresses a subtler risk unique to LLMs: the model may “remember” a famous crash from its training data.

3. Strategies under monitoring

Two deliberately **independent** strategies (daily return correlation ≈ 0.003) that break on *different* events — so a monitor that only works on one is exposed:

Strategy	Type	Characteristic break
AL PCA	Avellaneda–Lee statistical arbitrage (PCA-defactored mean reversion)	2007 quant meltdown
JT momentum	Jegadeesh–Titman cross-sectional momentum (long top decile / short bottom)	2009 momentum crash

Each strategy emits a daily PnL curve with a fixed schema (`Date`, `port_ret`, `equity`, `drawdown`) that both monitoring layers consume identically.

4. Classical monitoring layer (the baseline)

Four statistical change-point detectors run **continuously** over each daily PnL curve (so every window inherits genuine warm-up history):

- **Page–Hinkley** (cumulative-deviation test)
- **BOCPD** (Bayesian online change-point detection)
- **2-state HMM** (regime-switching; HMM parameters fit **out-of-sample** on pre-event data)
- **Distributional threshold** (rolling tail test)

Aggregation rule: an alarm fires when ≥ 2 detectors trigger within 5 days. This is the classical baseline against which the agentic layer is measured.

5. Agentic monitoring layer (the treatment)

A pipeline of LLM agents (local `qwen2.5:3b` via Ollama, structured-JSON outputs) fed the same point-in-time data:

```
Guardrails → Triage → News Context Agent → Performance Supervisor →
JSONL audit log
(as-of,      (skip / (reads filtered news, (telemetry + classical
timestamp,   cheap / emits risk flags +   alarms + news summary
lookahead)   thinking/ narrative)           → {state, action,
                                                root_cause, confidence}))
                                                escalate)
```

- **Guardrails** enforce information parity (§2) and fail closed.
- **Triage** decides how much model to spend per day: `skip` (quiet days), `thinking`, or `classical_escalation` — controlling inference cost.
- **News Context Agent** ingests the day's filtered news and emits structured risk signals.

- **Performance Supervisor (v3 prompt)** ingests strategy telemetry, the classical detector alarms, and the news summary, and emits a validated JSON assessment: `state ∈ {NORMAL, WATCH, ALERT}`, `action`, `root_cause`, `confidence`.
- Every model call is logged to JSONL with prompt version, raw output, validated assessment, and latency — for full auditability and blind qualitative scoring.

6. Multi-market design

To test **H2 (generalisation)** beyond a single market, the system was refactored behind a **market-provider abstraction** (`MarketProfile` + provider interfaces for universe, strategy PnL, news, macro, calendar). Each market plugs in its own data sources behind stable interfaces; the classical + agentic pipeline is shared and identical across markets. The US S&P 500 path is preserved **byte-identical** through the refactor (a golden-path regression test enforces this).

Market	Universe	News	Macro	Status
US S&P 500	PIT membership incl. delisted	FNSPID (11.8M articles)	FRED/ALFRED (rates, VIX)	Reference
Global energy	Curated ~34 megacaps (US- listed + ADR)	GDELT (point-in- time)	FRED commodity curves (WTI, Brent, Henry Hub)	Implemented
ASX 200	—	—	—	Deferred (survivorship-free data not freely available)

Energy-market design notes. Energy regime breaks are commodity-driven, so the macro provider carries oil/gas curves rather than only rates. Because energy is a *single sector*, the AL statistical-arbitrage engine is run in **PCA-defactoring mode** over the whole universe (PCA removes the common oil factor; the strategy trades the mean-reverting residuals) rather than the sector-ETF sleeve scheme used for the broad S&P 500. The curated fixed universe carries a documented, minor survivorship caveat (energy megacaps rarely delist; M&A exits are enumerated).

7. Evaluation design

Windows. Each market is evaluated on labelled slices of the timeline, split into a **development set** (used for calibration and prompt iteration — treated as contaminated) and a **blind confirmatory test set** (no parameter tuning or prompt changes after seeing outcomes; access gated by a freeze tag). US uses 4 event + 2 calm dev windows and 6 blind test windows. Energy uses windows within the news-data coverage range (e.g., the **2020 oil crash**, the **2022 energy spike**, plus calm controls). Each **event** window carries a ground-truth **onset** date; each **calm** window carries none (any alarm in a calm window is a false positive).

Metrics (computed per detector/agent, per window, and aggregated):

- **Detection latency** — trading days from onset to first alarm (H1: lower is better)
 - **False-positive rate** — alarms per day inside calm windows (H1: lower is better)
 - **Precision / recall** — on event detection
 - **Qualitative reasoning score** — blind rubric on agent root-cause narratives
-

8. Methodological controls (validity threats and mitigations)

Threat	Mitigation
Survivorship bias	PIT universe incl. delisted; gating test that a known crash is invisible on survivor-only data
Look-ahead in macro	ALFRED vintages (as-published values)
Look-ahead in news	Publication-timestamp filtering + fail-closed lookahead assertion
LLM pre-training leakage	Two-pass evaluation: Condition A (standard), Condition B (dates masked <code>XXXX-XX-XX</code> so the model cannot anchor on famous dates), Condition C (synthetic crash windows with fabricated year-2099 dates and no real event names). A leakage bound is computed: $\text{memorisation} \leq A - \min(B, C)$.
HMM fit on evaluation data	HMM parameters estimated out-of-sample on pre-event history
Cross-strategy confounding	Two statistically independent strategies ($\text{corr} \approx 0.003$)

9. Case study: the agentic layer on the 2020 oil crash

A worked demonstration on the **global-energy** market, window = 2020 oil crash (2020-02-14 → 2020-05-29), strategy = AL PCA. The agentic monitor ran with real `qwen2.5:3b` inference over 73 trading days (triage: 56 skip / 12 thinking / 5 escalation), producing **17 structured supervisor assessments**. The agent raised **ALERT** on **2020-03-10** (confidence 0.85) — one day after the crash onset (2020-03-09) — and held `WATCH` / `ALERT` through the turmoil.

Illustrative PnL overlay (not the primary experiment, which is detection metrics — this is a downstream what-if). Mapping the monitor's state to exposure (NORMAL → full, WATCH → half, ALERT → cash), applied with a **1-day lag (no look-ahead)**, over a \$1M

account for the window:

	Return	Max drawdown
Baseline (buy & hold energy)	−29.4%	−62.8%
Strategy (AL PCA)	+12.1%	−8.8%
Strategy + agentic monitor	+8.4%	−3.6%

The monitor **more than halved the strategy's max drawdown** (−8.8% → −3.6%) at a cost of ~3.7 pts of return — the risk-management trade a fast monitor is meant to deliver.

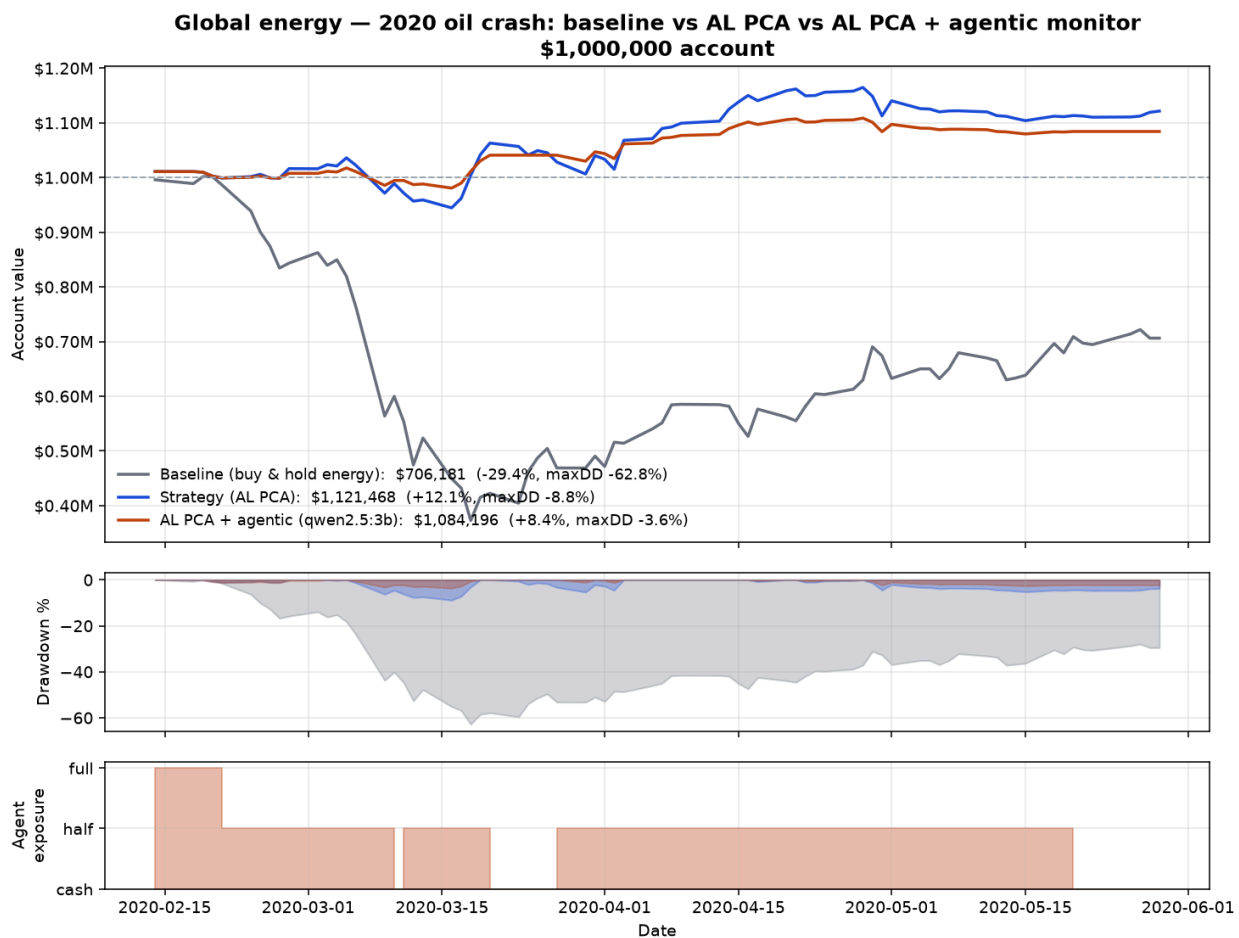


Figure 1. AL PCA stat-arb over the 2020 oil crash. The agentic monitor (orange) tracks the strategy but cuts exposure on WATCH/ALERT signals (bottom panel), reducing max drawdown from −8.8% to −3.6%.

The same procedure applied to **JT momentum** is more dramatic. Momentum spiked early (energy names dispersed at the crash open) then whipsawed down, ending roughly flat (+1.9%) with a –28.5% swing. The agent — which raised **ALERT** on the crash onset day itself (2020-03-09) — cut exposure after the run-up, **locking in the early gains and avoiding the whipsaw**, so it improved *both* return and drawdown:

(JT momentum, 2020 oil crash)	Return	Max drawdown
Baseline (buy & hold energy)	–29.4%	–62.8%
JT momentum	+1.9%	–28.5%
JT momentum + agentic	+13.7%	–9.8%

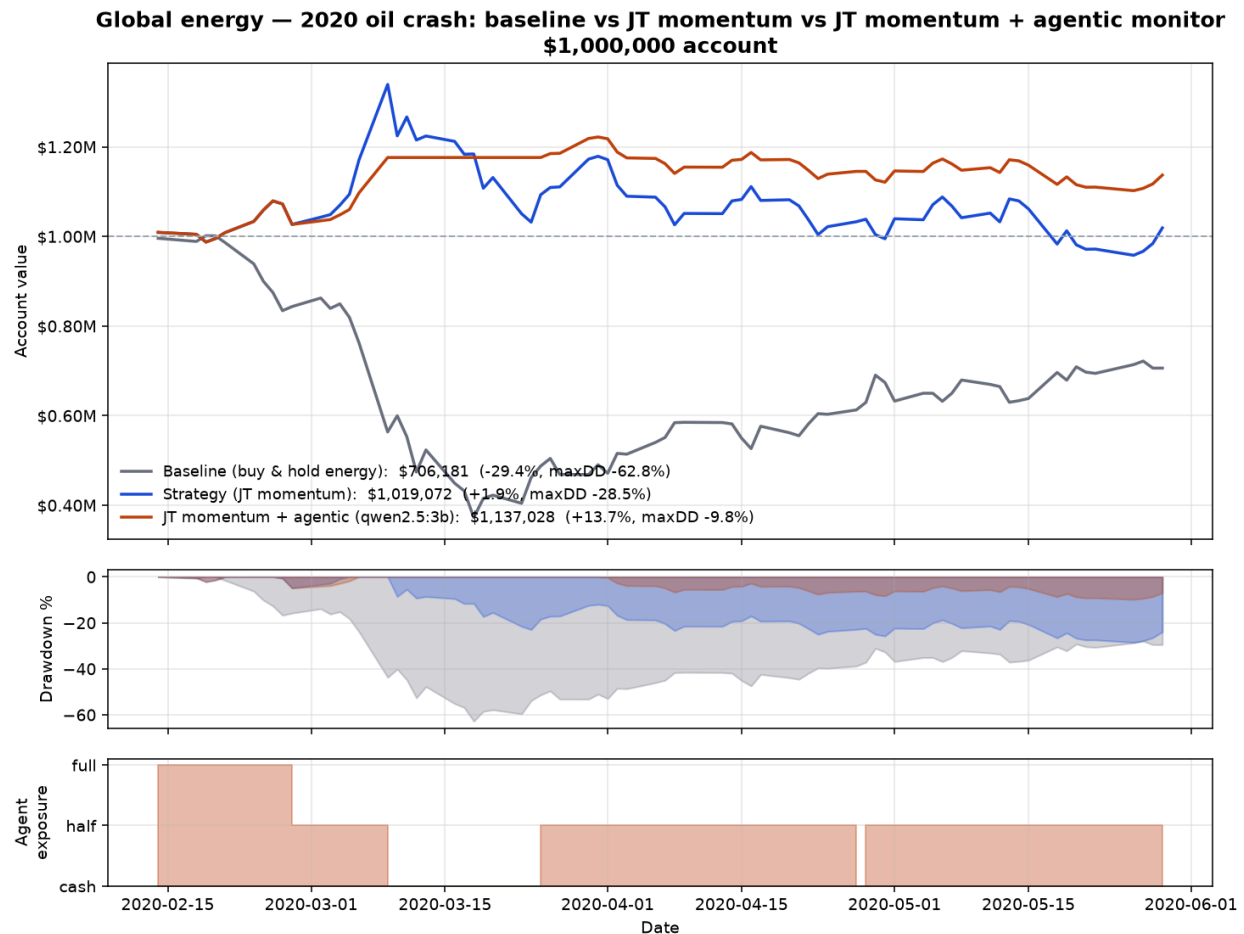


Figure 2. JT momentum over the 2020 oil crash. The monitor improved both return (+1.9% → +13.7%) and max drawdown (–28.5% → –9.8%).

This contrast directly illustrates H3 (the advantage differs by strategy): the monitor delivered pure *risk reduction* on the mean-neutral AL PCA book, but *both* higher return and lower risk on the whipsawed momentum book.

Framing for interpretation. The core hypotheses (H1–H3) are about **detection quality** (latency, FPR, precision/recall), not PnL. The overlay above is an intuitive illustration of *why* faster detection matters; it is not itself the claimed result, and its exposure-mapping rule is a modelling choice stated explicitly.

10. Reproducibility

All code is version-controlled. Representative commands (run from `monitoring/` in the project venv):

```
# Classical monitoring, either market
python run_classical.py --market sp500 --model stub
python run_classical.py --market energy --model stub

# Agentic monitoring, one window (real LLM)
python run_agentic.py --market energy --window oil_crash_2020 \
    --strategy AL_PCA --model ollama:qwen2.5:3b

# Pre-training leakage harness (Conditions A/B/C)
python run_leakage.py --strategy AL_PCA --model stub
```

Every agentic run emits a JSONL audit log (prompt version, raw + validated output, latency) enabling blind qualitative scoring and full replay. Strategy PnL curves, evaluation-window definitions with onsets, and the point-in-time data builders are all in the repository.

11. Scope and limitations (stated honestly)

- **ASX 200** is deferred: survivorship-free constituent + delisted-price data is not freely available, and the project's validity depends on that data being honest.
- The **energy universe** is a curated fixed set (not the full point-in-time index), with a documented minor survivorship caveat justified by energy-megacap stability.

- The **local** `qwen2.5:3b` model is a small, fast model chosen for cost and reproducibility; results characterise this model class, not the frontier.
- The **PnL overlay** (§9) is an illustration, not the primary metric.
- Energy news coverage (**GDELT DOC 2.0**) begins ~2017, constraining energy windows to that range; pre-2017 energy events require the GDELT GKG raw files.